

Figure 3: Results on *HalfCheetah* task with different imperfect expert setting. **Left**: Different number of state-action pairs; **Right**: Different level of imperfectness.

demonstrations of different qualities by mixing the demonstrated data from perfect (**Expert**) and imperfect (**Demo**) policies with different ratios. The **Right** of Figure 3 presents the results of our method and baselines with these demonstrations. It implies that the quality of demonstrations will significantly affect the performances of all the evaluated methods, and expert data with high quality can facilitate policy optimization to a certain extent. We can also see that our method overall outperforms the two counterparts even though the expert data becomes perfect (by setting the ratio to 1.00), indicating that our constraint-based method can exploit the expert data more efficiently than other methods based on penalty departures or pre-training.

5.4 Ablation Analysis II: Sensitivity to Constraint Tolerance

Now we will further investigate how can the design of the core soft constraint affect the performative results of our method. More specifically, we’re interested in the tolerance factor d . By varying the initial value of d and annealing strategies (namely, different annealing factor ϵ), we will explore the sensitivity of our algorithm regarding them.

Different tolerance. We design four groups of parameters for the ablation experiments on the tolerance choosing in *HalfCheetah* task, where the annealing mechanism is disabled by setting ϵ fixed at zero, and choose initial tolerance d_0 from $\{10^0, 10^{-1}, 10^{-3}, 10^{-6}\}$. The learning curves are plotted in **Left** of Figure 4. As the results demonstrate, when given relatively large tolerance, the exploration reference from demonstrations will not work as the constraint almost does not affect policy optimization. In contrast, a too-small tolerance will hurt the final performance when the demonstrations are imperfect. Therefore, hand-crafting the tolerance for the constraint can be difficult, and an automatic adjustment with the annealing mechanism should be adopted.

Fixed vs. Annealing tolerance. In the previous experiment, we mention the importance of annealing of tolerance. Now we explore the advantages of annealing mechanism quantitatively in *HalfCheetah* task. Since our annealing is to enlarge the tolerance along training, we only choose two not-too-large initial tolerances d_0 from $\{10^{-3}, 10^{-6}\}$, and select the annealing factor ϵ from $\{0, 2 \times 10^{-3}, 10^{-3}, 10^{-6}\}$. Corresponding learning curves are shown in **Right** of Figure 4. We can see that the performances of our method with an

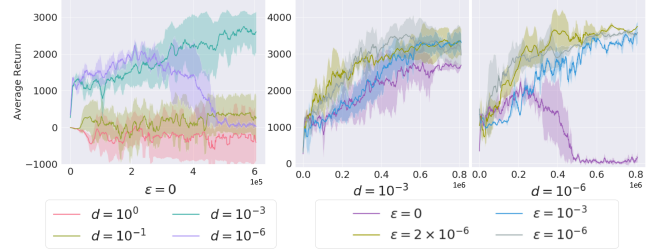


Figure 4: Learning curves over on *HalfCheetah* task. **Left**: ablation study about different tolerance factor d ; **Right**: sensitivity of choosing fixed or annealing strategy of tolerance.

annealing tolerance are overall better than with a fixed one (simply by setting ϵ as zero). Moreover, when the annealing factor ϵ is set properly, the performance of our method is not sensitive to the minor changes of ϵ as the results of different factors are almost at the same level. This further demonstrates the robustness of our proposed method.

6 Conclusion

In this paper, we investigate the problem of RLfD with imperfect expert data. Compared to existing RLfD problem setting, this new setting does not require the expert to be optimal, which can be more practical for real-world demonstrators like a human. We show that current penalty based RLfD methods will suffer from the issue of optimality and convergence when being applied to the setting of imperfect experts both theoretically and empirically. To this end, we propose to employ the expert data as a soft constraint and reformulate RLfD as a constrained policy optimization problem to narrow the negative impact of the imperfectness. We also provide an efficient learning algorithm for solving the challenging constrained optimization problem with scalable policy model like neural networks. Experiments on physical control benchmarks demonstrate the effectiveness of our proposed method over other RLfD counterparts. While we still assume the expert data to be collected from the same domain as the current conducted task, further exploration on combining our work with representation learning to enable learning with demonstrations across different domains could be a new direction of future work.

Acknowledgment

This research was funded by National Science Foundation of China (Grant No.91848206). It was also partially supported by the National Science Foundation of China (NSFC) and the German Research Foundation (DFG) in project Cross Modal Learning, NSFC 61621136008/DFG TRR-169. We would like to thank Dr. Boqing Gong and Dr. Tao Kong for their generous help and insightful advice.

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